

IDEATION PHASE

DEFINE PROBLEM STATEMENTS

Date	3rd October 2026
Team ID	6ab8c0aa8c7c66e70673078c
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	

2. IDEATION PHASE

The Ideation Phase focuses on identifying the access-control problem, understanding the needs of the users involved, and generating possible solutions before selecting the most suitable approach for implementation in ServiceNow.

2.1 DEFINE PROBLEM STATEMENTS

Problem Identification

The Institution Details (u_institution_details) table contains student information such as Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, and Description. The records belong to different branches, including ECE, EEE, and CSE.

A major requirement of the project is to ensure that users do not receive unrestricted access to all records and operations. Access must be controlled according to the user's assigned roles and, for Read access, the value of the Branch field.

The system therefore needs a mechanism to:

- Restrict unauthorized users from viewing records.
- Allow authorized users to view only the required records.
- Control Read, Create, Write, and Delete operations independently.
- Assign different permissions through custom roles.
- Use the Branch value to restrict record visibility.
- Prevent users from performing operations for which they do not have permission.
- Provide a way to verify the configured access rules.

Primary Problem Statement

How can access to student records in ServiceNow be controlled based on the user's assigned role and the Branch field, while providing separate permissions for Read, Create, Write, and Delete operations?

Specific Problem Statements

Problem 1 – Unauthorized Record Access

Users without the required permissions should not be able to access restricted student records.

Problem 2 – Branch-Based Record Restriction

The system needs to restrict Read access based on the Branch value, with the configured requirement allowing authorized users to view EEE branch records.

Problem 3 – Different Access Levels

Different users require different levels of access depending on their assigned roles.

Problem 4 – CRUD Control

Read, Create, Write, and Delete operations need to be controlled separately instead of providing unrestricted table access.

Problem 5 – Access Verification

The configured ACLs need to be tested with different users and role combinations to ensure that the intended restrictions are enforced.

Design Challenge

To design a ServiceNow ACL-based security mechanism that combines user roles and record field values to provide controlled access to the Institution Details table.

Desired Outcome

The desired solution should provide:

